EC442 Macroeconomics for MRes students
London School of Economics

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Problem Set 4 Solutions

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Growth Models with Human Capital

Question 1: Consider an economy in which the representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

The physical resource constraint is given by:

$$c_t + i_t = A k_t^{\alpha} h_{1,t}^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$\begin{aligned} k_{t+1} &= (1-\delta)k_t + i_t, \\ h_{t+1} &= B(h_{2,t})^{\gamma}, \end{aligned}$$

where $0 < \gamma < 1$. The allocation of human capital is subject to the constraint $h_t = h_{1,t} + h_{2,t}$, with $h_{1,t}, h_{2,t} \geq 0$, and the initial capital stocks $k_0 > 0$ and $h_0 > 0$ are given.

1. Define a sequential-markets equilibrium for this economy.

   Answer: A sequential markets equilibrium is a sequence of quantities $\left\{k_{t+1}^*, h_{t+1}^*, h_{1,t}^*, c_t^*\right\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ such that

   - Consumers maximize utility, taking prices as given: $\left\{k_{t+1}^*, h_{t+1}^*, h_{1,t}^*, c_t^*\right\}_{t=0}^{\infty}$ solves

$$\begin{aligned} \max_{\{k_{t+1}, h_{t+1}, h_{1,t}, c_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t \log(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} = (1-\delta)k_t + r_t k_t + w_t h_{1,t}, \\ & h_{t+1} = B(h_t - h_{1,t})^{\gamma}, \quad 0 \leq h_{1,t} \leq h_t \quad \text{for all } t \geq 0, \\ & k_0,\ h_0 \text{ given.} \end{aligned}$$

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- Firms maximize profits, taking prices as given: in every period, $\left(k_t^*, h_{1,t}^*\right)$ solves
$$\max_{k,h_1} \; Ak^\alpha h_1^{1-\alpha} - r_t k - w_t h_1.$$
- Markets clear: in every period, firms rent the physical capital and the human capital that consumers supply, and output is either consumed or invested:
$$c_t^* + k_{t+1}^* - (1-\delta)k_t^* = A(k_t^*)^\alpha (h_{1,t}^*)^{1-\alpha}.$$

2. Use first-order conditions to compute the steady state.

   Answer: The consumer's problem has the current-value Lagrangian
$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \log(c_t) + \lambda_{1,t}\big[(1-\delta+r_t)k_t + w_t h_{1,t} - c_t - k_{t+1}\big] + \lambda_{2,t}\big[B(h_t - h_{1,t})^\gamma - h_{t+1}\big] \Big\},$$
   where $\lambda_{1,t}$ and $\lambda_{2,t}$ are the multipliers on the budget constraint and on the law of motion for human capital, measured in units of period-t utility. The first-order conditions with respect to c_t, k_{t+1}, $h_{1,t}$, and h_{t+1}, followed by the firm's first-order conditions, are
$$\begin{aligned}
\frac{1}{c_t} - \lambda_{1,t} &= 0 \\
-\lambda_{1,t} + \beta(1-\delta+r_{t+1})\lambda_{1,t+1} &= 0 \\
\lambda_{1,t} w_t - \lambda_{2,t} B\gamma (h_t - h_{1,t})^{\gamma-1} &= 0 \\
-\lambda_{2,t} + \beta\lambda_{2,t+1} B\gamma (h_{t+1} - h_{1,t+1})^{\gamma-1} &= 0 \\
(1-\alpha) A k_t^\alpha (h_{1,t})^{-\alpha} - w_t &= 0 \\
\alpha A k_t^{\alpha-1} (h_{1,t})^{1-\alpha} - r_t &= 0
\end{aligned}$$
   which imply
$$\begin{aligned}
\frac{c_{t+1}}{c_t} &= \beta(1-\delta+r_{t+1}) \\
1-\delta+r_{t+1} &= \frac{w_{t+1}}{w_t}\gamma B h_{2,t}^{\gamma-1} \\
w_t &= (1-\alpha) A k_t^\alpha (h_{1,t})^{-\alpha} \\
r_t &= \alpha A k_t^{\alpha-1} (h_{1,t})^{1-\alpha},
\end{aligned}$$
   together with the law of motion for human capital and feasibility (consumption plus investment equals output):
$$\begin{aligned}
h_{t+1} &= B h_{2,t}^\gamma, \qquad h_t = h_{1,t} + h_{2,t}, \\
c_t + k_{t+1} &= (1-\delta)k_t + A k_t^\alpha (h_{1,t})^{1-\alpha}.
\end{aligned}$$

In a steady state, these conditions reduce to

$$\begin{aligned} 1 &= \beta(1-\delta+r) \\ 1-\delta+r &= \gamma B h_2^{\gamma-1} \\ w &= (1-\alpha)Ak^{\alpha}(h_1)^{-\alpha} \\ r &= \alpha A k^{\alpha-1}(h_1)^{1-\alpha}, \\ h &= Bh_2^{\gamma}, \qquad h = h_1 + h_2, \\ c+\delta k &= Ak^{\alpha}(h_1)^{1-\alpha}, \end{aligned}$$

such that

$$\begin{aligned} r &= \frac{1}{\beta} - 1 + \delta \\ h_2 &= (\beta\gamma B)^{\frac{1}{1-\gamma}} \\ h &= Bh_2^{\gamma} \\ h_1 &= h - h_2 \\ \frac{k}{h_1} &= \left(\frac{\frac{1}{\beta} - 1 + \delta}{\alpha A} \right)^{\frac{1}{\alpha-1}}, \\ c &= Ak^{\alpha}(h_1)^{1-\alpha} - \delta k, \end{aligned}$$

where h_2 follows from combining the first two conditions, k from h_1 and the ratio k/h_1, and w from the third condition.

3. Show that the conditions for the first welfare theorem are satisfied (implying that we can solve a social planning problem to compute the equilibrium allocation).

   Answer: The preferences of the agent satisfy local non-satiation since a marginally higher level of consumption always yields strictly higher utility. Furthermore, there are no externalities and the markets are complete. Hence, the First Welfare Theorem applies.

   Why does this mean that we can solve a social planning problem to compute the equilibrium allocation? The argument has three steps:

   - *A market equilibrium is a planner equilibrium.* By the first welfare theorem, any equilibrium allocation is Pareto efficient. With a representative consumer, this means that it solves the social planning problem.
   - *Uniqueness of planner equilibrium.* Utility is strictly concave and the technology is convex, so the social planning problem has a unique solution. Hence, there is at most one market equilibrium and if it exist it is the solution of the social planning problem.

- *Existence.* The first welfare theorem does not say that an equilibrium exists, but we can construct one from the solution of the social planning problem, by setting prices equal to marginal products, $r_t = \alpha A k_t^{\alpha-1} h_{1,t}^{1-\alpha}$ and $w_t = (1-\alpha) A k_t^{\alpha} h_{1,t}^{-\alpha}$. At these prices, firms choose the planner's inputs, since these satisfy the firm's first-order conditions and profits are concave in the inputs. The consumer chooses the planner's allocation, since it satisfies the budget constraint, the consumer's first-order conditions, and the transversality conditions, which are sufficient because the consumer's problem is concave. Finally, markets clear because the planner's allocation is feasible.

The second welfare theorem guarantees this in general, but possibly with transfers and under additional assumptions in an infinite-horizon economy (Stokey et al., 1989, Chapter 15). Here the direct verification is simpler, and no transfers are needed: the representative consumer owns all the capital and firms make zero profits. Finally, markets are complete here: there is no uncertainty, so the only commodities are the good at each date, and consumers can trade the good across dates by saving in physical capital (in equilibrium, the restriction $k_{t+1} \geq 0$ does not bind). One saving device is enough because each period there is a single state of the world; with uncertainty, markets would be complete only if there were a claim for each possible state of the world.

4. Formulate the Bellman equation corresponding to the social planning problem in this economy.

   Answer: The Bellman equation can be written as

$$v(k,h) = \max_{k',h'} \log \left((1-\delta)k + Ak^{\alpha} \left(h - \left(\frac{h'}{B} \right)^{\frac{1}{\gamma}} \right)^{1-\alpha} - k' \right) + \beta v\left(k', h'\right)$$

$$\text{s.t. } k' \geq 0, \;\; 0 \leq \left(\frac{h'}{B} \right)^{\frac{1}{\gamma}} \leq h.$$

5. Solve the value function numerically for the parameter values $\beta = 0.6$, $\alpha = 0.3$, $\delta = 0.6$, $\gamma = 0.6$, $A = 1$, $B = 2$ (these parameters approximate a calibration in which each model period corresponds to a decade). Limit the state space in each dimension to $[0.4\bar{k}, 1.5\bar{k}]$ and $[0.4\bar{h}, 1.5\bar{h}]$, where $\bar{k}$ and $\bar{h}$ are the steady state values. Provide graphs of the value and policy functions, and provide time-series graphs comparing outcomes over 10 periods in three economies: Economy A starts out in the steady state ($k_{a,0} = \bar{k}$, $h_{a,0} = \bar{h}$), Economy B starts out with the steady-state level of physical capital, but only half as much human capital ($k_{b,0} = \bar{k}$, $h_{b,0} = \bar{h}/2$), and Economy C starts out with the steady-state level of

human capital, but only half as much physical capital ($k_{c,0} = \bar{k}/2$, $h_{c,0} = \bar{h}$). Provide economic intuition why shortfalls of physical versus human capital lead to qualitatively different transition paths.

Answer: I show the value function and the policy functions in Figure 1, and Figure 2 shows the simulations of the economy starting out with different levels of physical and human capital. From Figure 2, we see that the economy moves much faster to the steady state after a deviation in physical capital than in human capital. Also, if we start out with human capital below its steady state level, the consumer chooses to decrease the physical capital stock too. The reverse is much less true: the human capital stock is almost not affected by variations in the physical capital stock.

Why is this the case? If capital stock is below the steady state, it is possible to compensate by allocating more human capital to the production of consumption/investment goods. However, if the human capital stock is below the steady state, it is not possible to compensate the decrease in human capital allocation by physical capital.

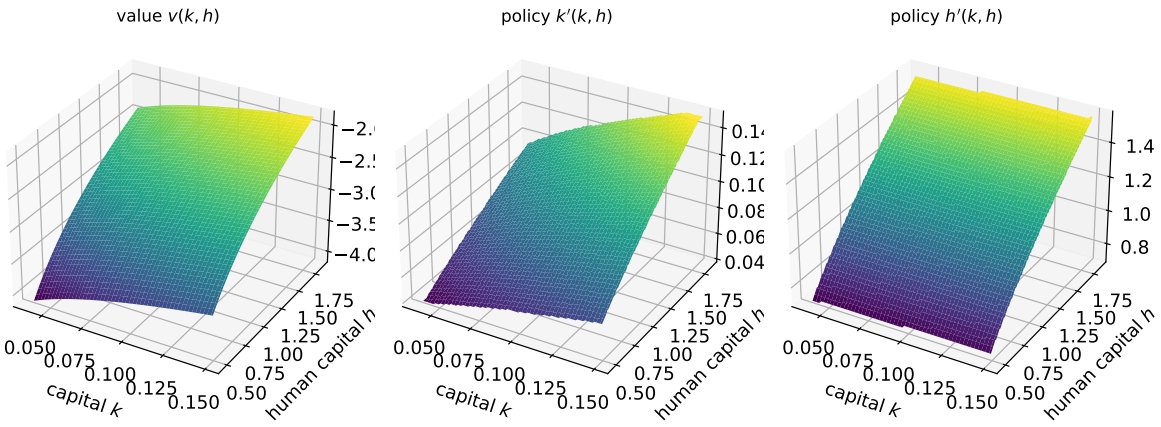


Figure 1: Value function and policy functions

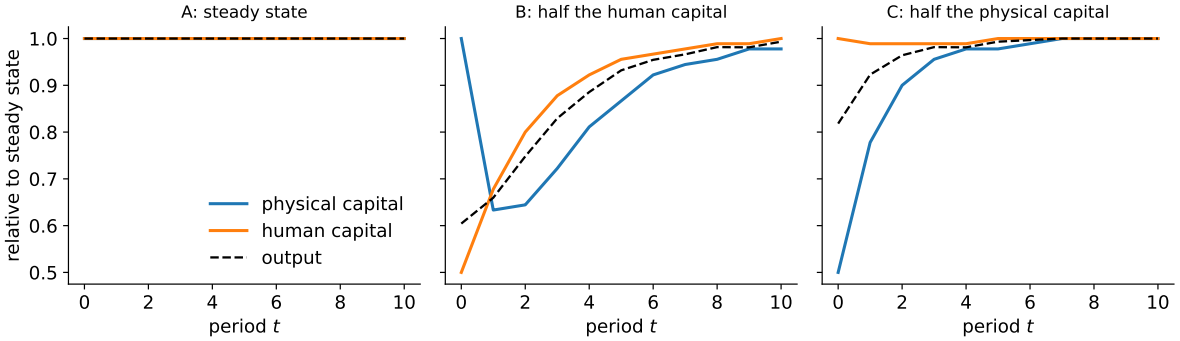


Figure 2: Time series of physical capital, human capital and output. All variables are expressed as a fraction of their steady state level.

Question 2: We now turn to a version of the Lucas (1988) model, in which human capital can be accumulated under constant returns. The representative consumer still has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

The physical resource constraint is given by:

$$c_t + i_t = A k_t^{\alpha} h_{1,t}^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$\begin{aligned} k_{t+1} &= (1-\delta)k_t + i_t, \\ h_{t+1} &= B h_{2,t}. \end{aligned}$$

The allocation of human capital is subject to the constraint $h_t = h_{1,t} + h_{2,t}$, with $h_{1,t}, h_{2,t} \geq 0$, and the initial capital stocks $k_0 > 0$ and $h_0 > 0$ are given.

1. Write down the social planning problem for this economy. Let $v^*(k,h)$ be the value function corresponding to the planning problem.

   Answer: Using $h_{2,t} = h_t - h_{1,t}$, the social planning problem is

$$\begin{aligned} v^{\star}(k,h) = \max_{\{c_t, k_{t+1}, h_{1,t}, h_{t+1}\}_{t=0}^{\infty}} & \sum_{t=0}^{\infty} \beta^t \log(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} = (1-\delta)k_t + A k_t^{\alpha} h_{1,t}^{1-\alpha}, \\ & h_{t+1} = B(h_t - h_{1,t}), \\ & c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq h_{1,t} \leq h_t \quad \text{for all } t \geq 0, \\ & k_0 = k, \quad h_0 = h. \end{aligned}$$

2. Use first-order conditions to compute the balanced growth path for the economy (in which both types of capital and consumption all grow at a common rate).

   Answer: The planner's problem has the current-value Lagrangian

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & \log(c_t) + \lambda_{1,t}\big[(1-\delta)k_t + A k_t^{\alpha} h_{1,t}^{1-\alpha} - c_t - k_{t+1}\big] \\ & + \lambda_{2,t}\big[B(h_t - h_{1,t}) - h_{t+1}\big] \Big\}, \end{aligned}$$

   where $\lambda_{1,t}$ and $\lambda_{2,t}$ are the multipliers on the resource constraint and on the law of motion for human capital, using $h_{2,t} = h_t - h_{1,t}$. The first-order conditions

with respect to c_t, k_{t+1}, $h_{1,t}$, and h_{t+1} are

$$\begin{aligned}
\frac{1}{c_t} - \lambda_{1,t} &= 0 \\
-\lambda_{1,t} + \beta\lambda_{1,t+1}\left(1 - \delta + \alpha A k_{t+1}^{\alpha-1} h_{1,t+1}^{1-\alpha}\right) &= 0 \\
\lambda_{1,t} w_t - \lambda_{2,t} B &= 0 \\
-\lambda_{2,t} + \beta\lambda_{2,t+1} B &= 0,
\end{aligned}$$

where $w_t \equiv (1-\alpha) A k_t^{\alpha} h_{1,t}^{-\alpha}$ is the marginal product of human capital in production. As in Question 1, the third condition gives $\lambda_{2,t} = \lambda_{1,t} w_t / B$, and substituting it into the fourth gives $\lambda_{1,t} w_t = \beta\lambda_{1,t+1} B w_{t+1}$. Together with the first two conditions, this implies

$$\begin{aligned}
\frac{c_{t+1}}{c_t} &= \beta\left(1 - \delta + \alpha A k_{t+1}^{\alpha-1} h_{1,t+1}^{1-\alpha}\right) \\
1 - \delta + \alpha A k_{t+1}^{\alpha-1} h_{1,t+1}^{1-\alpha} &= \frac{w_{t+1}}{w_t} B.
\end{aligned}$$

The second condition is again a no-arbitrage condition between physical and human capital: one unit of human capital diverted from production yields B units of human capital next period, each worth w_{t+1} in production.

Does this model have a steady state? In a steady state, c, k, h, and h_1 are constant, so that $c_{t+1} = c_t$ and $w_{t+1} = w_t$, and the two conditions require

$$\beta B = 1 \qquad \text{and} \qquad 1 - \delta + \alpha A \left(\frac{k}{h_1}\right)^{\alpha-1} = B.$$

Hence, if $\beta B \neq 1$, there is no steady state. The reason is that the return on investing in human capital, B, does not depend on how much human capital the economy has. In Question 1, the return $\gamma B h_2^{\gamma-1}$ falls as more human capital is invested, so that a unique level of investment equates it to the rate of time preference, $1/\beta$. Here there are no diminishing returns, so no level of capital can do that: if $\beta B > 1$, as with our parameters, the economy grows forever, and if $\beta B < 1$, it shrinks forever. In the knife-edge case $\beta B = 1$, there is a steady state, but not a unique one: the ratios are pinned down, $h_2/h = 1/B = \beta$ and k/h_1 by the condition above, but the level of human capital is not. Every level of h, with k in proportion, is a steady state, and the initial conditions determine which one the economy reaches. This is the balanced growth path below, with $g = \beta B = 1$.

On a balanced growth path, all variables grow at the same gross rate g. Then $k_t/h_{1,t}$ is constant, so the wage w_t is constant too, and the two conditions give

$$g = \beta B, \qquad 1 - \delta + \alpha A \left(\frac{k}{h_1}\right)^{\alpha-1} = B, \quad \text{that is,} \quad \frac{h_1}{k} = \left(\frac{B - 1 + \delta}{\alpha A}\right)^{\frac{1}{1-\alpha}}.$$

Finally, the law of motion $h_{t+1} = Bh_{2,t}$ gives $h_2/h = g/B = \beta$, so that $h_1 = (1-\beta)h$, and on a balanced growth path

$$\frac{h}{k} = \left(\frac{B-1+\delta}{\alpha A}\right)^{\frac{1}{1-\alpha}} \frac{1}{1-\beta}.$$

3. Make a direct argument to show that:

$$v^\star(\lambda k, \lambda h) = \frac{1}{1-\beta}\log(\lambda) + v^\star(k,h).$$

Answer: Let $F(k,h,k',h') \equiv (1-\delta)k + Ak^\alpha \left(h - h'/B\right)^{1-\alpha} - k'$ denote consumption. By definition,

$$\begin{aligned} v^\star(\lambda k, \lambda h) = \max_{\{k_{t+1},h_{t+1}\}_{t=0}^\infty} & \sum_{t=0}^\infty \beta^t \log F(k_t,h_t,k_{t+1},h_{t+1}) \\ \text{s.t.} \quad & k_{t+1} \geq 0, \quad 0 \leq h_{t+1} \leq Bh_t \quad \text{for all } t \geq 0, \\ & k_0 = \lambda k, \quad h_0 = \lambda h. \end{aligned}$$

The function F is homogeneous of degree one, and so are the constraints: a sequence $\{k_{t+1},h_{t+1}\}_{t=0}^\infty$ is feasible from $(\lambda k, \lambda h)$ if and only if $\{\tilde{k}_{t+1},\tilde{h}_{t+1}\}_{t=0}^\infty$, with $\tilde{x}_t \equiv x_t/\lambda$, is feasible from (k,h). Using this change of variables,

$$\begin{aligned} v^\star(\lambda k, \lambda h) &= \max_{\{\tilde{k}_{t+1},\tilde{h}_{t+1}\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t \log F(\lambda\tilde{k}_t, \lambda\tilde{h}_t, \lambda\tilde{k}_{t+1}, \lambda\tilde{h}_{t+1}) \\ &= \max_{\{\tilde{k}_{t+1},\tilde{h}_{t+1}\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t \left(\log(\lambda) + \log F(\tilde{k}_t, \tilde{h}_t, \tilde{k}_{t+1}, \tilde{h}_{t+1})\right) \\ &= \frac{1}{1-\beta}\log(\lambda) + v^\star(k,h), \end{aligned}$$

where the maximizations are over sequences feasible from (k,h).

4. Formulate the Bellman equation corresponding to the social planning problem in this economy.

Answer: The Bellman equation is

$$\begin{aligned} v(k,h) = \max_{k',h'} & \log\left((1-\delta)k + Ak^\alpha\left(h - \frac{h'}{B}\right)^{1-\alpha} - k'\right) + \beta v\left(k',h'\right) \\ \text{s.t. } & k' \geq 0, \;\; 0 \leq \frac{h'}{B} \leq h. \end{aligned}$$

5. Solve the value function numerically for the parameter values $\beta = 0.6$, $\alpha = 0.3$, $\delta = 0.6$, $A = 1$, $B = 2$. You can make use of the previous result, which implies that the solution v of the Bellman equation, which coincides with $v^\star$, satisfies:

$$v(k,h) = \frac{1}{1-\beta}\log(k) + v\left(1, \frac{h}{k}\right).$$

That is, you only need to compute the one-dimensional value function:

$$\tilde{v}\left(\frac{h}{k}\right) = v\left(1, \frac{h}{k}\right).$$

Limit the state space to the interval $[0.4\bar{h}/\bar{k}, 2\bar{h}/\bar{k}]$, where $\bar{h}/\bar{k}$ is the ratio of human to physical capital on the balanced growth path. Provide graphs of the value and policy functions (i.e., three-dimensional surfaces for varying k and h), and provide time-series graphs comparing outcomes over 10 periods for the same three economies as before: Economy A starts out on the balanced growth path ($h_0/k_0 = \bar{h}/\bar{k}$), Economy B starts out with the same level of physical capital as Economy A, but only half as much human capital, and Economy C starts out with as much human capital as Economy A, but only half as much physical capital. I recommend plotting the time series in logs, so that balanced growth shows up as straight lines.

Answer: The one-dimensional value function $v\left(1, \frac{h}{k}\right)$ can be written as

$$\begin{aligned}
v\left(1, \frac{h}{k}\right) = &\max_{k',h'} \left\{ \log\left(1 - \delta + A\left(\frac{h}{k} - \frac{h'}{B}\right)^{1-\alpha} - k'\right) + \beta v\left(k', h'\right) \right\} \\
&\text{s.t. } 0 \le k' \le A\left(\frac{h}{k} - \frac{h'}{B}\right)^{1-\alpha} + 1 - \delta, \;\; 0 \le h' \le B\frac{h}{k} \\
= &\max_{k',h'} \left\{ \log\left(1 - \delta + A\left(\frac{h}{k} - \frac{h'}{B}\right)^{1-\alpha} - k'\right) \right. \\
&\qquad \left. + \frac{\beta}{1-\beta}\log(k') + \beta v\left(1, \frac{h'}{k'}\right) \right\} \\
&\text{s.t. } 0 \le k' \le A\left(\frac{h}{k} - \frac{h'}{B}\right)^{1-\alpha} + 1 - \delta, \;\; 0 \le h' \le B\frac{h}{k}
\end{aligned}$$

or, equivalently, if we define $z' \equiv \frac{h'}{k'}$, then

$$v\left(1, \frac{h}{k}\right) = \max_{z'} \left\{ \max_{k'} \left\{ \log\left(1 - \delta + A\left(\frac{h}{k} - \frac{z'k'}{B}\right)^{1-\alpha} - k'\right) + \frac{\beta}{1-\beta}\log(k') \right\} + \beta v\left(1, z'\right) \right\}$$
$$\text{s.t. } 0 \leq k' \leq A\left(\frac{h}{k} - \frac{z'k'}{B}\right)^{1-\alpha} + 1 - \delta, \;\; 0 \leq z'k' \leq B\frac{h}{k}.$$

This suggests an algorithm to update the value function:

- for each $\frac{h'}{k'}$ on the grid, maximize current value function over k' given constraints
- then choose $\frac{h'}{k'}$ that yields maximum value
- update value function

The resulting value function and policy functions are shown in Figure 3. To go from h/k to (k, h), we use the homogeneity of the value function, $v(k, h) = \frac{1}{1-\beta}\log(k) + \tilde{v}(h/k)$, and of the policy functions, which are homogeneous of degree one. In proportional terms, the value function increases more strongly in human capital than in physical capital: on the balanced growth path, $\partial v / \partial \log(h) \approx 2.1$ and $\partial v / \partial \log(k) \approx 0.4$, which add up to $1/(1-\beta) = 2.5$.

Additional remarks

The Lucas model and externalities. We just solved the social planning problem of a simple version of the Lucas (1988) model of human capital accumulation, and as in Question 1 by the first welfare theorem its solution is also the competitive equilibrium. However, the original paper is mostly about externalities: there, output also depends on the *average* level of human capital in the economy, and individuals don't take into account that they all determine the average level when deciding how much to invest in human capital. With this externality, the first welfare theorem fails: the competitive equilibrium no longer solves the social planning problem, and it has too little investment in human capital and a lower growth rate than the efficient allocation, so that subsidizing education can raise growth and welfare.

References

Robert E. Lucas, Jr. On the mechanics of economic development. *Journal of Monetary Economics*, 22(1):3–42, 1988.

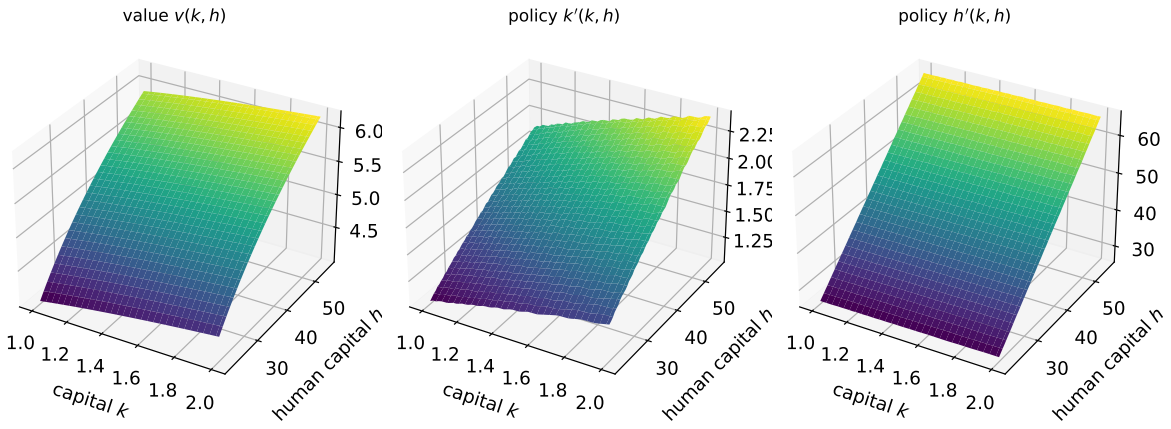


Figure 3: Value function and policy functions

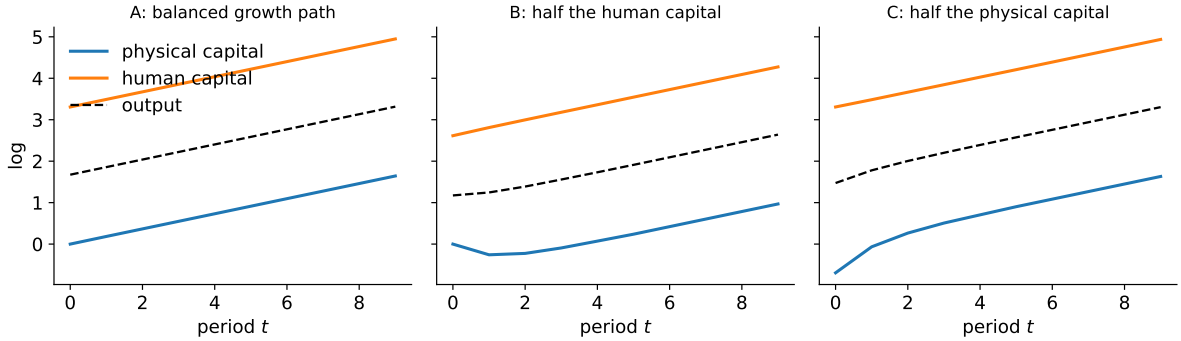


Figure 4: Simulations of the economy (in logs), starting out with different levels of human and physical capital.

Nancy L. Stokey, Robert E. Lucas, Jr., and Edward C. Prescott. *Recursive Methods in Economic Dynamics*. Harvard University Press, Cambridge, MA, 1989.